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ORIGINAL PAPER

Micromechanical analysis of cohesive granular materials using the discrete element method with an adhesive elasto-plastic contact model

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Abstract An adhesive elasto-plastic contact model for the discrete element method with three dimensional non-spherical particles is proposed and investigated to achieve quantitative prediction of cohesive powder flowability. Simulations have been performed for uniaxial consolidation followed by unconfined compression to failure using this model. The model has been shown to be capable of predicting the experimental flow function (unconfined compressive strength vs. the prior consolidation stress) for a limestone powder which has been selected as a reference solid in the Europe wide PARDEM research network. Contact plasticity in the model is shown to affect the flowability significantly and is thus essential for producing satisfactory computations of the behaviour of a cohesive granular material. The model predicts a linear relationship between a normalized unconfined compressive strength and the product of coordination number and solid fraction. This linear relationship is in line with the Rumpf model for the tensile strength of particulate agglomerate. Even when the contact adhesion is forced to remain constant, the increasing unconfined strength arising from stress consolidation is still predicted, which has its origin in the contact plasticity leading to microstructural evolution of the coordination number. The filled porosity is predicted to increase as the contact adhesion increases. Under confined compression, the porosity reduces more gradually for the load-dependent adhesion compared to constant adhesion. It was found that the contribution of adhesive force to the limiting friction has a significant effect on the bulk unconfined strength. The results provide new insights and propose a micromechanical based measure for characterising the strength and flowability of cohesive granular materials.

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List of symbols

d	Particle diameter (m)
d_{avg}	Average particle diameter (m)
e	Co-efficient of restitution
g	Gravitational constant (m/s^2)
m^*	Equivalent mass of the particles (kg)
N	Number of particles
n	Non-linear index parameter
P	Pressure (kPa)
$\boldsymbol{u}$	Unit normal vector
Z	Coordination number at peak
Zi	Instantaneous coordination number
F_{at}	Average adhesive strength at contact (N)
f_0	Constant adhesive strength at first contact (N)
f_{t}	Contact tangential force (N)
f_{ct}	Coulomb limiting tangential force (N)
f_{nd}	Normal damping force (N)
f_i^c	Contact force (N)
f_{atp}	Average tensile force at the peak (N)
f_{ts}	Tangential spring force (N)
f_{td}	Tangential damping force (N)
f_{hys}	Hysteretic spring force (N)

$f_{ts(n-1)}$	Tangential spring force at previous time step (N)
I_i	Moment of inertia (m^4)
l_i^c	Vector from centre of particle to the contact point
k_1	Loading stiffness parameter (kN/m)
k_2	Unloading/reloading stiffness parameter (kN/m)
k_{adh}	Adhesive stiffness parameter (kN/m)
k_t	Tangential stiffness (kN/m)
$\boldsymbol{v}_t$	Relative tangential velocity (m/s)
v_n	Magnitude of relative normal velocity (m/s)
σ_u	Unconfined yield strength (kPa)
σ_a	Axial stress (kPa)
σ_t	Bulk tensile strength (kPa)
σ_1	Axial consolidation stress (kPa)
ρ	Particle density (kg/m^3)
ε	Total bulk deformation
ε_a	Bulk axial strain
ε_p	Total plastic deformation
β_n	Normal dashpot co-efficient
β_t	Tangential dashpot coefficient
ϕ	Angle of friction (°)
Δf_{ts}	Incremental tangential force (N)
δ	Total normal overlap (m)
δ_{max}	Maximum normal overlap (m)
δ_p	Plastic overlap (m)
η	Sample bulk porosity
η_c	Consolidated bulk porosity
η_f	Fill porosity
μ	Co-efficient of friction
μ_r	Coefficient of rolling friction
τ_i	Total applied torque (N m)
ω_i	Unit angular velocity vector (radian/s)
λ_p	Contact plasticity
λ_b	Bulk plasticity
$\dot{\gamma}$	Shear rate

1 Introduction

Many powders and particulate solids are stored and handled in large quantities in various industries such as pharmaceutical, food, and chemical industries: they constitute over 75 % of material feedstock in industry [1]. One of the common issues experienced by many of these materials is the flow difficulty arising from material cohesion. Poor flow affects storage, transfer, production, packing, compaction, fluidisation, distribution, and end use of the product in a negative manner, resulting in, for example, arching and ratholing in silo storage, segregation in blending, under- and over-dosage in filling, which ultimately lead to economic losses.

The flowability of cohesive solids is often measured using the flow function introduced by Jenike [2], which describes the unconfined strength as a function of the consolidation stress. The flow function of a cohesive solid is an important material property for appropriate, efficient, and economic design of bulk handling equipment. Apart from the more traditional direct shear tests such as the Jenike circular cell [2] and the Carr–Walker [3] or Schulze annular ring cell [4], indirect uniaxial shear tests are often used to evaluate the flowability for a cohesive material [5–11]. The flow function of a cohesive solid shows the manifestation of the unconfined yield strength that arises from the historical consolidation stress and therefore the cohesive strength is stress-history dependent. Such stress-history dependent cohesive behaviour must be captured if a numerical model is to successfully simulate the cohesive powder flow. This paper describes the development of a phenomenological discrete element method (DEM) model coupled with a calibration methodology for quantitative prediction of such powder flow behaviour.

1.1 DEM contact models for adhesion

A number of contact models [12–21] have been proposed for modelling cohesive powders. However, the common adhesion models including JKR [14], DMT [12], and Matuttis and Schinner [21], which are elastic contact models, may not be able to capture the stress history dependent behaviour shown in experiments of cohesive powders [5–11].

Since the contact area between two fine particles is very small, even moderate forces (in the order of 1 nN) can cause plastic deformation at the contact [22] which can give rise to a stress history dependent behaviour. Therefore, it is proposed that contact plasticity has an important role in correctly simulating the stress history dependency. Walton and Braun [23] were probably the first to introduce a normal force displacement (NFD) elasto-plastic bilinear spring model, based on an approximation of finite element analysis (FEA) results, to account for the plasticity at the contact for cohesionless solids. Ning and Thornton [24] also presented an NFD contact model for elasto-plastic spheres, in which they used a normal force displacement relationship that follows the non-linear Hertzian relationship [25] until the contact yields and thereafter assumed a linear model. For unloading after plastic deformation the NFD contact model again follows the normal force-displacement relationship proposed by Hertz [22], but uses a larger effective radius allowing for contact enlargement resulting from irreversible plastic deformation.

Based on a FEA study on spherical particles, Vu-Quoc and Zhang [26] found that after the initial plastic yielding, the NFD contact model is neither linear as assumed by Thornton [27] nor following the Hertzian relationship. They argued that Thornton's NFD contact model produces a softer force-displacement relationship and a smaller coefficient of restitution for the same maximum normal displacement level compared to the FEA results. Vu-Quoc and Zhang [26] proposed an NFD contact model with nonlinear unloading stiffness which is independent of the applied load.

The first NFD contact model with elasto-plastic deformation and adhesion was introduced by Thornton and Ning [16]. For cohesive solids, the plastic deformation in the contact region causes a larger effective radius of the deformed contact region, and upon unloading a larger resultant pull-off force is observed. They modelled such behaviour using a modified JKR curve with a larger contact radius. Walton and Johnson [28] argued that the modified JKR model has a complicated formulation and may have difficulty to meld with friction and rolling resistance. A more elaborate and detailed contact model accounting for load, time and rate dependent visco-elastic, plastic, visco-plastic, adhesion and dissipative behaviour was proposed by Tomas [29]. In the contact model initial loading response is Hertzian which switches to linear elasto-plastic loading after yielding. Contact unloading is elastic and follows Hertzian contact. The contact model also takes account of hysteretic behaviour during the unloading/reloading cycle. A linear adhesion limit accounts for increasing adhesion with increasing plastic deformation. However this model is computationally intensive. By ignoring the initial non-linear portion before yielding on the basis that fine particles show a negligible range of Hertz-like behaviour [30], Tykhoniuk et al. [31] showed that simpler piece-wise linear models such as the ones proposed in [32] produced similar results at a smaller computational cost. Luding et al. [15,33] further improved the piece-wise linear model by recognizing the loading history dependent nature of the unloading/reloading stiffness by making the unloading/reloading stiffness a function of the previously experienced maximum overlap. This has introduced an element of nonlinearity in unloading/reloading path. Walton and Johnson [28] also proposed a contact model similar to Tomas and Luding's model but separated the rate of increase of the pull-off force from the slope of the tensile force-displacement unloading curve which requires an additional parameter to describe the adhesive behaviour.

1.2 DEM studies of uniaxial test

A number of DEM studies have been conducted to investigate the behaviour of cohesive solids using the above mentioned contact models. Moreno-Atanasio et al. [34] conducted DEM simulations of a uniaxial test using JKR elastic adhesive contact model to investigate the effect of cohesion on the flowability of a polydisperse particulate system. The mechanical properties of glass beads were used as DEM input parameters. They argued that for consolidation stresses in the range of 0–10 kPa the model produced unconfined yield strengths that could be classified as highly cohesive according to Jenike's classification. The unconfined strength increased with increasing consolidation stress showing stress history dependence. It may be argued that the stress history dependence is because the cohesive powder forms a loose initial structure which collapses on the application of load resulting in particle re-arrangement into a denser packing so that the deformation does not recover significantly on the removal of load, especially at low stress levels. The use of an elastic contact model may be suitable for simulating some materials such as elastic glass beads at low stresses where deformation is mainly due to particle re-arrangement. However, for many industrial processes, the stress can be much higher and the materials are evidently not elastic. In the case of cohesive solids, an adhesive elastic contact model may not be able to represent the realistic behaviour, where the permanent plastic deformation gives rise to stress history dependent behaviour.

Hassanpour and Ghadiri [35] conducted both experimental and DEM studies on the flowability of powders using indentation and uniaxial compression tests on a small assembly of powders compacted at low pressures. They adopted a contact model based on the Hertzian analysis for the elastic regime, Thornton and Ning's [16] model for the plastic deformation and the JKR theory [14] for the adhesion force. DEM simulations based on single particle mechanical properties showed that the unconfined yield strength varies linearly with the consolidation stress, which is similar to the experiment, but there is a poor quantitative agreement between experiment and simulation, with a discrepancy up to 160 %. The discrepancy was attributed to rough estimation of the adhesion parameter and the adoption of spherical particle shape in the DEM simulation.

The great majority of DEM studies in the literature were conducted using either 2D disks or 3D spheres. A number of numerical and experimental studies [36–39] have shown that particle shape affects the flowability of the powder. While spherical particles with higher rolling friction are introduced to simulate particle shape by many researchers (e.g. [40]), it has been reported that spherical particles cannot represent the "real" solids regardless of the angle of inter-particle friction [41]. The spherical particles fails to capture interlocking related dilation, voidage distribution, and material shear strength arising from interlocking [41]. Therefore, it is important to introduce an appropriate degree of non-sphericity in particle shape to capture the behaviour of real solids which are very rarely spherical. This study deploys 3D non-spherical particle using multi-sphere technique as described in [42] which is also being used by many others [43–47].

1.3 Overview and focus of study

In this paper, the development of an adhesive elasto-plastic DEM contact model for cohesive powders is presented. The model comprises a nonlinear hysteretic spring model to account for the elastic–plastic contact deformation and an adhesive force component that is a function of the plastic contact deformation. The proposed contact model is conceptually similar to existing models [15,17,18] but has the additional capability of non-linearity for the loading and unloading paths. Atomic Force Microscopy (AFM) measurements of the contact force displacement curve of very fine particles of fumed silica and titania have shown smooth non-linear behaviour during both loading and unloading [48]. The non-linear behaviour after plastic yielding has also been reported by Vu-Quoc and Zhang [26]. In order to model both linear and nonlinear NFD behaviour of real solids, a power law is proposed for both loading and unloading paths in the contact model.

The predictive capability of the proposed DEM contact model is evaluated by attempting to simulate the behaviour of a limestone powder under quasi-static and slow shearing regimes. The study explores the full spectrum of cohesive behaviour from filling of a space (fill porosity) to loading under confined compression, and finally unconfined loading to failure. A key focus is to produce a quantitative model for the unconfined strength as a function of the consolidation stress. The calibration of the model parameters employs experimental data of bulk solid characteristics, recognising the differences between the model and the real solids at the particle level. The details of the model and computational methodology are described in the following section, followed by modelling strategy and the DEM implementation to simulate confined and unconfined uniaxial loading to failure. The DEM predictions are compared with the experimental results before the effects of various parameters on filled porosity, compressibility, and unconfined strength are explored. The causes of the stress history dependency of the unconfined strength of cohesive solids are then investigated. Finally, the effect of limiting frictional criteria on the unconfined strength is explored.

2 Computational methodology

2.1 Equations of motion

The discrete element method (DEM) was first developed by Cundall and Strack [49] as a tool for analysing quasi-static problems related to densely packed granular materials. Interaction is modelled using the soft contact approach where rigid particles are allowed to overlap each other at the contact point with small overlaps, typically less than 1 % of the particle diameter at each time step. The contact force is calculated according to the contact model as a function of the overlap.

The changes in positions and velocities of the particles due to the contact and gravitational forces are calculated from the integration of Newton's motion equations. For particle i the translational motion equation is

$$m_i \frac{d^2}{dt^2}\boldsymbol{x}_i = \boldsymbol{f}_i + m_i \boldsymbol{g} \tag{1}$$

where m_i is the mass of the particle, t is time, $\boldsymbol{x}_i$ is its position, $\boldsymbol{f}_i$ is the summation of all forces acting on the particle ($\boldsymbol{f_i} = \sum_c \boldsymbol{f_i^c}$), and $\boldsymbol{g}$ is acceleration due to gravity. The rotational motion equation for particle i is given as:

$$I_i \frac{d}{dt}\boldsymbol{\omega_i} = \boldsymbol{T_i} \tag{2}$$

where I_i is the moment of inertia for particle i, $\boldsymbol{\omega_i}$ is its angular velocity and $\boldsymbol{T_i}$ is the total torque acting on it, which is defined by Eq. 3, where $\boldsymbol{l_i^c}$ is the vector from the centre of particle i to the contact point and $\boldsymbol{f_i^c}$ is the contact force.

$$\boldsymbol{T_i} = \sum_{i=1}^{Nc} \boldsymbol{l_i^c} \times \boldsymbol{f_i^c} \tag{3}$$

A sufficiently small integration time step is required to ensure the stability of simulation by having a sufficient number of time steps within each collision. The time step is usually calculated as a percentage of the Rayleigh time step [50–52], typically less than 10 %.

2.2 Visco-elasto-plastic adhesive frictional contact model

The DEM contact model proposed here is based on the physical phenomena observed in adhesive contact between micron sized particles or small agglomerates [53]. When two particles or agglomerates are pressed together, they undergo elastic and plastic deformations. It is assumed that the pull-off (adhesive) strength increases with an increase of the plastic contact area. A non-linear contact model that accounts for both the elastic–plastic contact deformation and the contact-area dependent adhesion is proposed. The schematic diagrams of particle contact and normal force-overlap ($f_n - \delta$) curve for this model are shown in Fig. 1a.

The loading, unloading, re-loading, and adhesive branches are characterised by five parameters: the virgin loading stiffness parameter k_1, the unloading and reloading stiffness parameter k_2, the constant adhesive strength f_o, the adhesive stiffness parameter k_{adh} and the stiffness exponent n. During the initial loading, the contact model follows the virgin loading path, k_1, upon unloading of the contact; the contact will switch from the virgin loading path to the unloading/reloading path, k_2. During reloading, the contact force initially follows along the reloading k_2 path but switches to

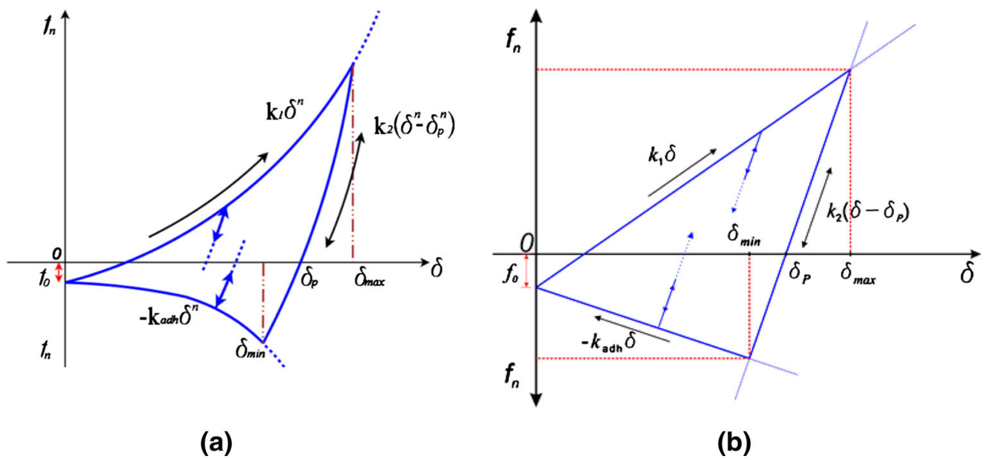

Fig. 1 Normal contact force-displacement function for the contact model **a** non-linear **b** linear

the virgin loading k_1 path when the previous maximum loading force is reached. Unloading along the k_2 path below the plastic overlap δ_p results in the development of an adhesive force until the maximum adhesive force is reached at $-k_{adh}\delta^n_{min} + f_0$. Further unloading past this point results in a reduction in both the normal overlap and the adhesive force until separation occurs ($\delta = 0$). If reloading of the contact occurs while on the adhesion branch, the contact will reload along a k_2 path (there is an infinite number of k_2 paths depending on the point of first unloading), until the virgin loading k_1 path is reached, and will continue loading along k_1 path on further increase of load. The contact information is lost when the particles are separated. This model does not consider hysteretic behaviour during reloading/unloading of the contact below δ_{min} as in Tomas's model. If k_1 is set equal to k_2 the model is reduced to an elastic contact model.

The unloading/reloading stiffness k_2 is not load dependant as in Luding's model [15]. The model has been implemented with a power law exponent parameter n to describe the shape of all the three loading-unloading branches—they all become linear when $n = 1$ (see Fig. 1b). Both linear and nonlinear trends for normal force displacement relationship have been reported in experiments (e.g. [48]) and numerical studies (e.g. [26]). In this first study of the model, we focus on studying the influence of the contact plasticity and adhesion parameters (k_2, k_{adh} and f_o) without invoking the nonlinearity by setting $n = 1$ which reverts the contact model to a piece-wise linear version that is conceptually similar to several existing models [15,28]. The simulations conducted using the non-linear model can be found in [54].

The contact model as described above was implemented through the API in EDEM® v2.4 (and subsequent versions), a commercial DEM code developed by DEM Solutions Ltd [55]. The total contact normal force, $\boldsymbol{f_n}$, is the sum of the hysteretic spring force, f_{hys}, and the normal damping force, f_{nd}:

$$\boldsymbol{f_n} = (f_{hys} + f_{nd})\boldsymbol{u}, \tag{4}$$

where, $\boldsymbol{u}$ is the unit normal vector pointing from the contact point to the particle centre. The force-overlap relationship for normal contact, $f_{hys,}$ is mathematically expressed by Eq. 5.

$$f_{hys} = \begin{cases} f_0 + k_1\delta^n & if \quad k_2(\delta^n - \delta_p^n) \geq k_1\delta^n \\ f_0 + k_2(\delta^n - \delta_p^n) & if \quad k_1\delta^n > k_2(\delta^n - \delta_p^n) > -k_{adh}\delta^n \\ f_0 - k_{adh}\delta^n & if \quad -k_{adh}\delta^n \geq k_2(\delta^n - \delta_p^n) \end{cases} \tag{5}$$

The normal damping force, $f_{nd,}$ is given by:

$$f_{nd} = \beta_n v_n \tag{6}$$

where v_n is the magnitude of the relative normal velocity, and β_n is the normal dashpot co-efficient expressed as:

$$\beta_n = \sqrt{\frac{4m^*k_1}{1 + \left(\frac{\pi}{\text{In } e}\right)^2}} \tag{7}$$

with the equivalent mass of the particles $m^* = (m_i m_j / m_i + m_j)$, where m is the mass of the respective particle, and the coefficient of restitution e given to the simulation code as an input parameter.

The contact tangential force, $\boldsymbol{f_t}$, is given by the sum of tangential spring force, $\boldsymbol{f_{ts}}$, and tangential damping force, $\boldsymbol{f_{td.}}$, as given by:

$$\boldsymbol{f_t} = (\boldsymbol{f_{ts}} + \boldsymbol{f_{td}}). \tag{8}$$

The tangential spring force is expressed in incremental terms:

$$\boldsymbol{f_{ts}} = \boldsymbol{f_{ts(n\text{-}1)}} + \boldsymbol{\Delta f_{ts}}, \tag{9}$$

where $\boldsymbol{f_{ts(n\text{-}1)}}$ is the tangential spring force at the previous time step, and $\boldsymbol{\Delta f_{ts}}$ is the increment of the tangential force and is given by:

$$\boldsymbol{\Delta f_{ts}} = -k_t\boldsymbol{\delta_t}, \tag{10}$$

where k_t is the tangential stiffness, and $\boldsymbol{\delta_t}$ is the increment of the tangential displacement. While varying values for the tangential stiffness have been used in the literature, in this study it is set as $2/7k_1$ [23].The tangential damping force is product of tangential dashpot coefficient, β_t, and the relative tangential velocity, $\boldsymbol{v_t}$, as given by Eq. (11):

$$\boldsymbol{f_{td}} = -\beta_t \boldsymbol{v_t}. \tag{11}$$

The dashpot coefficient β_t is given by:

$$\beta_t = \sqrt{\frac{4m^* k_t}{1 + \left(\frac{\pi}{\ln e}\right)^2}} \tag{12}$$

The limiting tangential friction force is calculated using the Coulombic friction criterion with the normal force modified by the adhesives forces given by:

$$f_{ct} \leq \mu \left(|f_{hys} + k_{adh}\delta^n - f_o| \right) \tag{13}$$

where f_{ct} is the limiting tangential force, and μ is the friction coefficient.

The default EDEM rolling friction model is adopted in this study. The total applied torque, $\boldsymbol{\tau_i}$, is given by:

$$\boldsymbol{\tau_i} = -\mu_r |f_{hys}| R_i \boldsymbol{\omega_i}, \tag{14}$$

where μ_r is the coefficient of rolling friction, R_i is the distance from the contact point to the particle centre of mass and $\boldsymbol{\omega_i}$ is the unit angular velocity of the object at the contact point.

There is some limited evidence in support of Eq. 13 in the experimental results from Skinner and Gane [56], who conducted micro-friction experiments between soft metal stylus and a hard smooth surface of graphite or diamond in a scanning electron microscope and found that the attractive force can be considered as an additive term to the normal force in calculation of limiting friction. Savkoor and Briggs [57], and Thornton and Yin [58] derived similar equation based on contact mechanics theory. The limiting friction criteria is consistent with the criteria set in other existing tangential force models [15,17,28,58].

The literature is rather vague about the tangential force models. One of the questions to be addressed in the area of tribology is whether the relationship between friction and normal force is linear. A number of microscopic interparticle friction experiments have reported the relationship between friction and normal force [56,59–64]. Both nonlinear (Hertzian and JKR) and linear (Coulombic) relationships were found between the frictional and normal forces. Briscoe and Kremnitzer [59] found non-linear relationship at tensile loads and linear relationship at compressive loads. They assumed a single asperity contact. In contrast, Ruths et al. [61] found a linear dependence in a study conducted on boundary friction of two different atomic silane monolayer with low adhesion in a single asperity contact with surface forces apparatus (SFA) and frictional force microscopy (FFM). Schwarz et al. [63] conducted a study using frictional force spectroscopy on nano-size carbon compounds with a single asperity and found that the frictional force was proportional to the 2/3 power of load similar to the theoretical model based on a Hertzian-type tip–sample contact. Friction force experiments by Jones et al. [60] found non-linear relationship between the normal and friction forces for ballotini indicating single asperity contacts and linear relationship for materials including alumina, limestone powder, zeolite, and titania supporting multi-asperity contact.

Although both linear and nonlinear relationships have been found between normal load and friction in microscopic friction experiments, we have assumed a linear relationship in this study for two main reasons. Firstly, real materials generally have multi asperity contacts and the literature suggests linear relationship for multi asperity or plastic contacts [60]. Secondly, elasto-plastic behaviour for fine adhesive particles can be expected in most cases of industrial processing and handling where averaged macroscopic stresses are above 1 kPa [22], and the nonlinear behaviour observed at low stresses in FFM can be ignored.

3 Modelling strategy and experimental characterisation

The above contact model in its full generic form captures the key elements of the frictional-adhesive contact mechanics in that: f_0 provides the van der Waals type pull-off strength; k_1 and k_2 provide the elastic–plastic contact; k_{adh} provides the adhesion unloading stiffness; the exponent n provides nonlinearity and the resulting contact plasticity defines the total contact adhesion (see Fig. 1a). The model is thus expected to be capable of modelling truly micron sized particles to study phenomena such as fine agglomeration, attrition and flow.

To apply the model at the single particle level requires the model parameters to be determined at the true particle-particle interaction level. This would require one to consider the enormous complexity of interfacial interaction including the influence of surface topology and chemistry and properties of the interstitial media etc. Whilst many studies have been reported on these measurements using microscopic measurement techniques such as AFM, Nano-indentation etc., the measurements tend to be on either highly idealised particle (such as specially manufactured perfect sphere) or suffer from enormous scatter and uncertainty with regard to the accuracy of the measurement [31,65]. Additionally, it is prohibitive to model each and every individual particle and cohesion arising from several different phenomena including van der Waals, capillary bridge and electrostatic forces separately, even in a very small system of fine powders. For example, a uniaxial test simulation of a cylindrical sample of 40 mm diameter and 80 mm in height with 4.7 μm sized limestone powder containing $>10^{12}$ particles may take in the order of 60 years if this was to be simulated with a 4 core, 64-bit computer.

This study focuses on an intermediate scale between the micro- and macro-scales, aiming to produce a phenomenological contact model that can reproduce the bulk cohesive strength, stress history dependency, and other behaviour evidenced in experiments. This study deploys 3D non-spherical particles in conjunction with a calibration strategy recognising the differences between the model and the real solids at the particle level to reproduce the bulk granular friction of a particulate system. The calibration of a mesoscopic DEM contact model requires experimental measurements characterizing the mechanical behaviour of powders. The mechanical behaviour of cohesive powders can be carefully measured using element tests such as biaxial test, true triaxial and hollow cylinder tests. However in practice these tests are expensive and slow to conduct and are almost never performed for many industrial applications requiring material characterisation. Here we have investigated a simpler technique that could be used for filling this important gap with the focus of providing test data for model calibration and simulation validation.

In this study, the Edinburgh Powder Tester (EPT) [5] is employed. The EPT is a semi-automated uniaxial tester, providing rapid measurements of various bulk mechanical properties of powders, inc- the stress–strain and the stress–porosity response during confined compression as well as the stress–strain response during unconfined compression including the peak unconfined strength. The reproducibility of results has been assessed for various materials with a coefficient of variation of typically less than 7 % [45,54].

In an EPT test, the sample is poured into the consolidation cylinder of diameter 40 mm and height 80 mm. The sample is loaded by a weight and the force is recorded by a load cell attached to the consolidation plunger. To minimise the effect of the friction between the particles and boundaries [6], the sample is allowed to compress from both the top and bottom in the EPT. After the sample is loaded for a selected consolidation time, the consolidation plunger is automatically retracted and the mould is manually slid down the pedestal, exposing a free standing column of consolidated powder sample. The unconfined sample is loaded to failure by a motor driven test piston at a speed of 0.4 mm/s, which was so chosen to conduct the test rapidly without affecting the measured unconfined yield strength. Watanabe and Groves [66] found that the unconfined strength of detergent samples was unaffected when the piston speed varied from 0.084 to 0.43 mm/s. The unconfined strength is automatically recorded, as well as the whole load-displacement curve. Consolidation stresses in the range of 16–96 kPa were applied in this study with 1 min consolidation time. ESKAL 500® (*KSL Staubtechnik GmbH*), a limestone powder with particles of 4.7 µm mean diameter was tested in this study (Fig. 2).

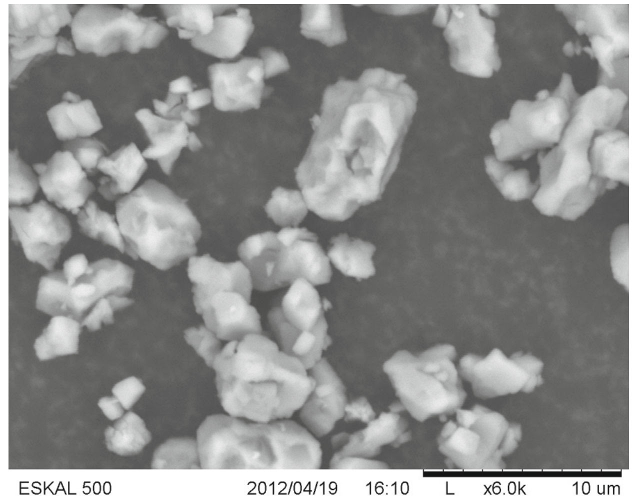

Fig. 2 Scanning electron microscopic (SEM) image of ESKAL 500

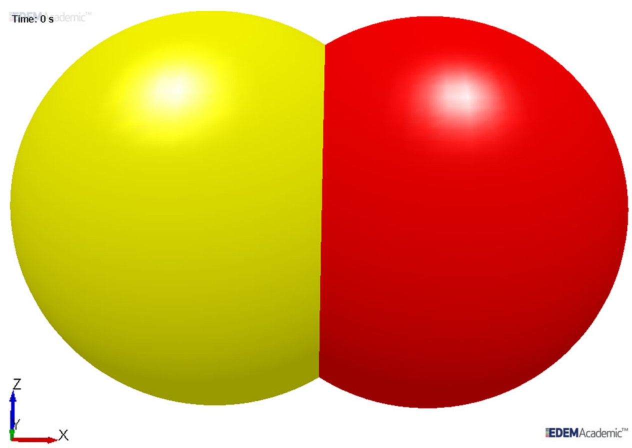

Fig. 3 Paired particle of aspect ratio = 1.5

4 Numerical simulation set-up

The DEM was used to simulate a series of uniaxial test experiments in a cylindrical mould. The proposed contact model was only applied to particle-particle interactions. Particle-geometry interactions were modelled using the Hertz-Mindlin (no-slip) contact model and hence no particle-geometry adhesion was allowed. Non-spherical particles were used, each consisting of two overlapping (paired) spheres of 1 mm diameter (d) giving a particle aspect ratio of 1.5 (Fig. 3). This relatively simple two-sphere shape was chosen based on the findings from two studies. The first study [67] showed that accurate representation is not necessary to produce satisfactory predictions as long as there is sufficient shape interlocking to generate the bulk friction. The second study showed that for purely spherical system, the maximum bulk friction under direct shear testing saturates at ~0.8 even for sliding friction of up to 2.0, whereas for two overlapping sphere with aspect ratio of 1.5, a full spectrum of bulk fric-

Table 1 Simulation parameters

Number of particles	2,200
Particle density, ρ (kg/m^3)	10,000
Co-efficient of restitution, e	0.4
Loading spring stiffness, k_1 (kN/m)	1
Unloading spring stiffness, k_2 (kN/m)	1, 1.25, 2, 5, 10, 100
Adhesive strength at first contact or constant adhesive strength, f_0, (N)	-0.002 to -0.05
Adhesive parameter stiffness, k_{adh} (kN/m)	0.1–100
Particle static friction, μ_{sf}	0.5
Particle rolling friction, μ_{rf}	0.001
Wall friction, μ_{wf}	0
Top and bottom platen friction, μ_{Pf}	0.1
Wall shear modulus, N/m^2	10^{10}
Poisson's ratio for particle to particle interaction	0.3
Poisson's ratio for particle to wall interaction	0.25
Simulation time-step (s)	8×10^{-7} to 2×10^{-6}

tion up to ~1.9 can be predicted [68]. Thus, by choosing the two-sphere model, we can expect the DEM model to be able to capture a wide range of bulk frictional characteristics for any real complex shape particles.

The parameters used in the simulations are listed in Table 1. The loading stiffness k_1 was chosen so that it is sufficiently stiff to avoid excessive particle overlap but not too stiff to avoid significant computational cost, and yet provide a close match to the experimental loading response. In order to reduce the computational time further, density scaling [52] was used as simulations are in quasi-static regime. The significance of dynamic effects in granular material is often evaluated by inertial number. The inertial number is used to describe the relative importance of inertia and confining stresses [69]:

$$I = \frac{\dot{\gamma} d_{avg}}{\sqrt{P/\rho}} \tag{15}$$

where $\dot{\gamma}$ = shear rate, d_{avg} = average particle diameter, P = pressure, and ρ = particle density.

An inertial number of less than 1×10^{-4} was observed for all simulation parameters and confirms that the simulations are in the quasi-static regime [69]. The simulation time step was chosen to be equal to $0.1\sqrt{m/k_2}$: no noticeable difference in results was found between simulations with time step of $0.3\sqrt{m/k_2}$ and $0.1\sqrt{m/k_2}$. The ratio of k_2/k_1 was varied from 1 to 100 by increasing k_2. Since the constant adhesive strength f_0 and the load-dependent adhesion represent different origins leading to cohesion, they are studied separately in Sect. 6 for the former and 7 for the latter.

DEM simulations using the proposed contact model were conducted for a series of uniaxial compression tests in a cylindrical mould of 15 mm diameter with top and bottom platens. This represents a scale of approximately 1/3 of the original experimental set-up, to reduce the computational cost. The initial filling height varied with DEM input parameters. However, the consolidated aspect ratio of the sample for the DEM simulations was kept in a narrow range of 1.2–1.4 which was used in the experiment. Each simulation consists of three stages—filling the cylindrical mould to form the initial packing used for all stress levels; confined consolidation to the required stress level and subsequent unloading; and finally unconfined compression of the sample to failure after the removal of the mould. The process is visualised in Fig. 4.

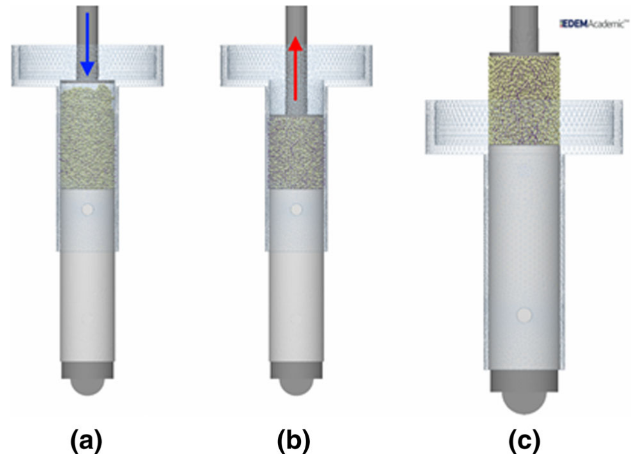

Fig. 4 DEM simulation of uniaxial test: **a** loading, **b** unloading, and **c** unconfined compression

The random rainfall method was adopted to form a random packing. To ensure that the system reached a quasi-static state, loading only commenced when the kinetic to potential energy ratio was less than 10^{-5} with a constant coordination number. The potential energy in the system is calculated based on a datum level of $z_d = 0$mm, which in this study relates the bottom of the mould. The confined consolidation process was conducted by moving the top platen downwards at a constant speed of 5 mm/s (strain rate $\approx 0.2\mathrm{s}^{-1}$) to apply a vertical compression. After consolidating the sample to the desired stress, the load on the assembly was released by moving the top platen upwards at the same constant speed. The lateral confining walls were then removed and the unconfined sample was allowed to relax for 0.1 seconds. This allowed the kinetic energy generated from the removal of the confining wall and upward retreat of the top platen to dissipate. The sample was then crushed to failure by moving the top platen downwards again, at a constant rate of 2 mm/s (strain rate $\approx 0.1\,\mathrm{s}^{-1}$). To investigate the effect of applied strain rate, simulations were conducted with varying strain rate in a range of 0.02–60 s^{-1} for elastic ($k_2 = k_1$) and elasto-plastic ($k_2 = 100k_1$) case. For both cases, it was found that confined and unconfined stress-strain behaviour does not change significantly with strain rate below $0.5\mathrm{s}^{-1}$. Therefore, strain rate smaller than $0.5\mathrm{s}^{-1}$ was chosen in this study. The failure of the sample was characterised by a drop in stress accompanied by a drop in the coordination number (Z) (see Fig. 15). The bottom platen remained stationary in all stages.

5 Numerical repeatability and prediction of flowability

Three numerical samples (samples 1, 2 and 3 below) with the same model parameters were created using the random particle generator implemented within the particle factory in the commercial code. These samples were then consolidated to 100 kPa prior to an unconfined compression test to failure to assess the variability in the results related to the generation of different particle packings in the assemblies. The scatter of these numerical samples was evaluated. Figures 5 and 6 present the axial stress-strain response during confined and unconfined compression, respectively. For confined compression, some variations in the stress-strain behaviour are noted. For unconfined compression, the average unconfined strength was found to be 6.6 kPa with a coefficient of variation (COV) = 3.4 %. The small COV indicates that randomly generated particle assembly has minor effect on the bulk response but is not independent of random assembly creation. While three data points would not usually be considered sufficient for rigorous statistical analysis, the very low COV indicates that the numerical scatter introduced by the random initial packing is small. The presence of numerical scatter should always be checked and used in the interpretation of the numerical results.

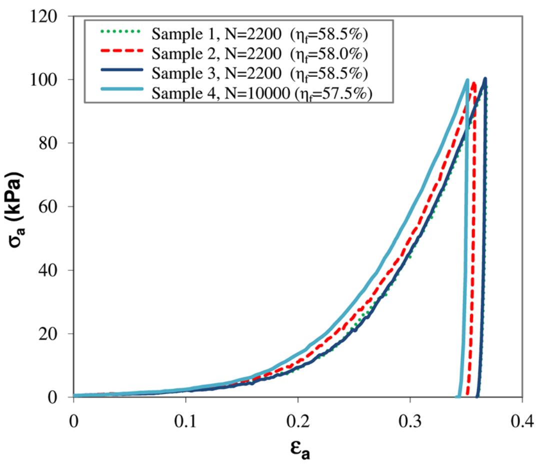

Fig. 5 Confined compression-axial stress (σ_a) versus axial strain (ε_a) for three random simulations and one large size simulation

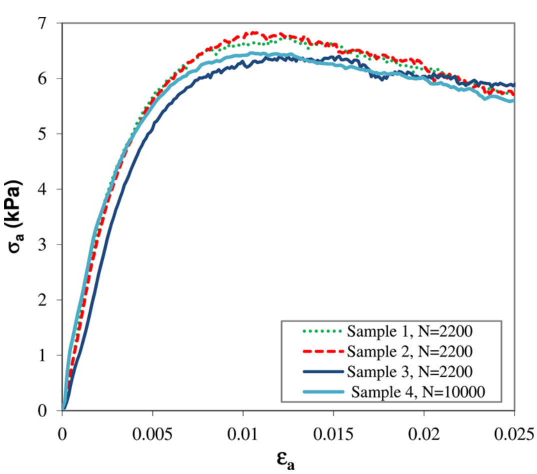

Fig. 6 Unconfined compression-axial stress (σ_a) versus axial strain (ε_a) for three random simulations and one large size simulation. Contact model parameters used are: $f_0 = -0.002\,\mathrm{N}$, $k_2 = 100$ and $k_{adh} = 0\,\mathrm{kN/m}$ (Note: N = number of particles, η_f = porosity corresponding to 0.5 kPa stress)

To investigate the influence of numerical sample size, an additional simulation (sample 4) with 10,000 paired-sphere particles with same particle aspect ratio (1.5) and other DEM input parameters as the simulation with 2,200 particles was performed. The number of particles were increased by increasing the diameter of the sample but keeping the sample aspect ratio same as the simulation with 2,200 particles. The results are included in Figs. 5 and 6. The comparison shows that the sample size does not have a significant effect on the prediction. For confined compression, the 10,000 particle system predicted ∼1.5 % smaller compression at the peak compared to the average peak strain of the samples with 2,200 particles. This can be attributed to a lower filled porosity for the 10,000 particle system because of a smaller boundary effect. For unconfined compression to failure, the unconfined peak strength is also within the scatter of the measurement for 2,200 particles. A more rigorous assessment of sample size would require more repeat random simulations to establish the statistical scatter which is beyond the scope of this paper.

The capability of the model is explored for predicting the unconfined yield strength of ESKAL 500 (a PARDEM reference solid) under different consolidation stresses. Figure 7 shows the predicted axial stress-strain responses during unconfined uniaxial compression of samples which have been consolidated at five stress levels: 16, 36, 56, 76, and 96 kPa. The initial loading stiffness increases as the consolidation stress increases. This has arisen from the change in the packing structure with decreasing porosity as consolidation stress increases. The maximum stress during unconfined compression (i.e. the unconfined strength σ_u) is plotted against the consolidation stress (see Fig. 8). For the parameters chosen, the model predicted a flow function that is in good quantitative agreement (within ∼12 %) with the experimental results. The simulation results show a strong dependence on the consolidation stress history as measured in the experiments. The reason for this stress-history dependency is related to the contact plasticity: the micromechanical aspects will be elucidated in the following section. Simulations were

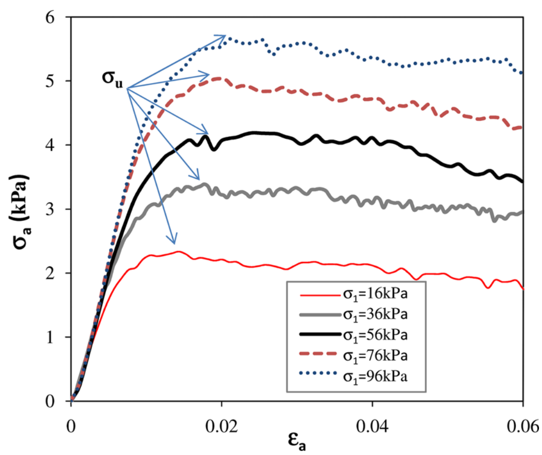

Fig. 7 Predicted unconfined axial stress (σ_a)-strain (ε_a) relationship. Contact model parameters used are: $f_0 = -0.003$ N, $k_2 = 3.5$ and $k_{adh} = 0$ kN/m

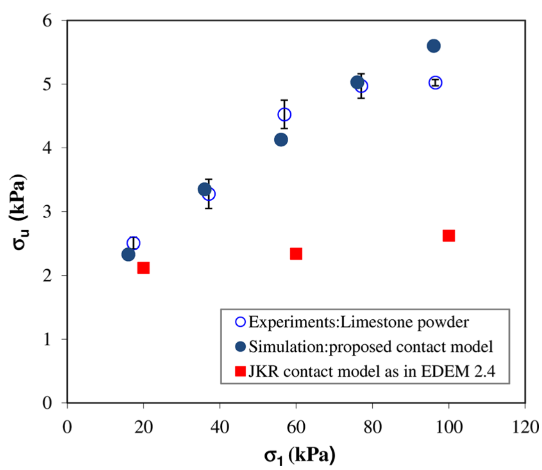

Fig. 8 Predicted versus test flow function for a limestone powder. Contact model parameters used are: $f_0 = -0.003$ N, $k_2 = 3.5$ and $k_{adh} = 0$ kN/m

also conducted using the modified JKR contact model with an elastic Hertzian contact in the EDEM code version 2.4 [70]. The surface energy of 1 J/m^2 and particle shear modulus of 10^{7}N/m^2 were used with other parameters kept the same as in Table 1. Figure 8 also shows the simulation results using the modified JKR cohesive model. The results show only a slight increase in unconfined strength as consolidation stress increases, giving an increasingly large discrepancy with the experimental observations with increasing consolidation stress when the JKR model is used. The results shown in Figs. 7 and 8 indicate that the implemented model is superior at capturing the salient features of a real cohesive powder.

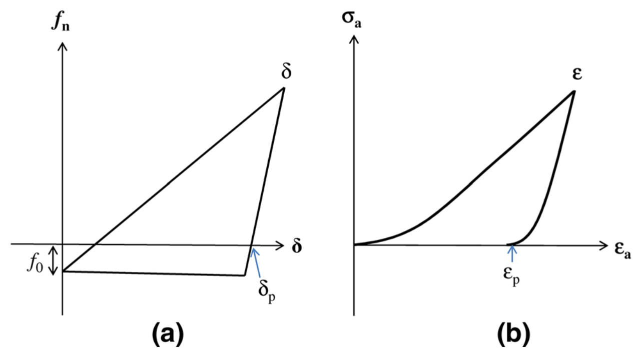

Fig. 9 **a** Linear elasto-plastic contact model with constant contact adhesion **b** typical bulk stress strain response during confined compression

6 Micromechanical analyses of porosity, plasticity and cohesion

The origin of the stress history dependent cohesion strength predicted by the model is explored here from a micromechanical point of view. The relationships between bulk material properties; namely the unconfined strength, the bulk plasticity and the porosity; and the microstructural properties are explored. Simulations of confined followed by unconfined compression, as described in Sect. 3, were performed for particle stiffness k_2 varying from 1 to 100 kN/m as listed in Table 1, and consolidation stress levels from 16 to 96 kPa. The level of contact plasticity was changed prior to the generation of the particles and was maintained for both the confined consolidation and unconfined compression to failure. The adhesion stiffness parameter k_{adh} was set to zero in the first instance so that the influence of the constant pull-off force f_0 can be explored (see Fig. 9a).

In an elasto-plastic contact, the contact plasticity λ_p may be defined as the ratio of the maximum plastic deformation δ_p to the total deformation δ at the contact. For the linear version of the contact model (n = 1, see Fig. 9a), λ_p becomes a simple function of the loading k_1 and unloading k_2 stiffnesses:

$$\lambda_p = \frac{\delta_p}{\delta} = 1 - \frac{k_1}{k_2} \tag{16}$$

Similarly the bulk plasticity (λ_b) can be defined as the ratio of the bulk plastic deformation ε_p to the total deformation ε, as shown in Fig. 9b.

$$\lambda_b = \frac{\varepsilon_p}{\varepsilon} \tag{17}$$

Figure 10 shows the predicted flow functions for a range of contact plasticity (induced by varying the stiffness ratio k_1/k_2). The slope of the flow function is an indication of the level of cohesion for a given set of parameters. The level of cohesion can increase from two sources: increasing coordination number and flattening of the contact under loading. It is evident that the flow function is strongly depen-

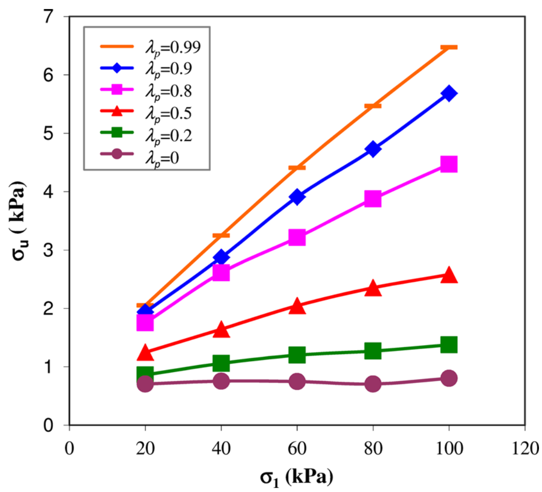

Fig. 10 Effect of particle contact plasticity on flow function ($f_0 = -0.002$ N, $K_{adh} = 0$ kN/m)

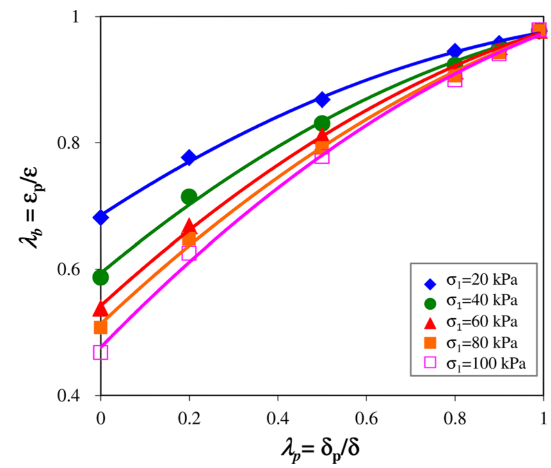

Fig. 11 Bulk plasticity (ε_p / ε) as a function of particle plasticity

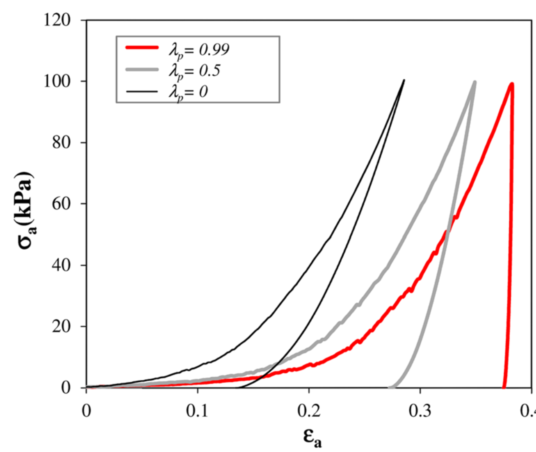

Fig. 12 Confined compression-axial stress (σ_a) versus axial strain (ε_a) with $\lambda_p = 0$

dent on the contact plasticity: a cohesive material with a constant particle-level adhesion force f_0 can change from being only slightly cohesive to moderately cohesive when the contact plasticity λ_p increases from 0 (elastic-no residual inter-particle contact deformation after unloading) to nearly 1 (no recovery of elastic deformation after unloading). When the contacts are elastic, the stress-history dependency largely disappears. Increasing the level of contact plasticity, which reduces elastic rebound, allows the consolidated assembly to maintain a lower consolidated porosity following unloading of the sample. The lower consolidated porosity in-turn leads to a higher number of inter-particle contacts, which generates a higher unconfined strength at which the assembly fails. Since load dependent adhesion was intentionally set as zero ($k_{adh} = 0$: see Fig. 9a), the increasing level of cohesion with increasing contact plasticity relates solely to the increasing contacts between particles.

The loss of stress-history dependent unconfined strength when contacts are elastic explains why adhesive models with elastic contact such as JKR and DMT models have difficulties in adequately capturing the stress history effect of cohesive solids. Most fine cohesive powders exhibit plasticity even at relatively low consolidation stress where a modest amount of force may lead to plastic yielding and irreversible deformation at the tiny particle-particle contact [71]. As shown by Hietsland [72], a modest amount of plastic deformation at particle contact may cause a dramatic increase in the strength of consolidated powders, which is in line with the simulation results in this study. From the mesoscopic perspective where a DEM particle represents a local assembly of primary particles, it is easy to see the presence of contact plasticity. An appropriate level of plasticity in the contact model should be included when modelling cohesive powders.

Figure 11 shows the effect of particle contact plasticity λ_p on bulk plasticity λ_b. It is seen that the bulk plasticity increases with increasing λ_p. When λ_p is small, the bulk plasticity arises predominantly from particle rearrangement which is greater during initial compression from the loose filled state; this give rise to a larger bulk plasticity for a smaller consolidation stress. As λ_p increases and contributes to overall deformation, the bulk plasticity increases as expected, but this must approach unity as the contact plasticity λ_p approaches unity. On its own, bulk plasticity cannot fully explain the stress history effect of a material because particles with elastic contacts can have very different bulk plasticity (see Fig. 11) but very little stress history effect (see Fig. 10), whilst particles with elastic–plastic contacts can have similar bulk plasticity but very strong stress history dependence.

Figure 12 shows the effect of contact plasticity λ_p on the loading-unloading response under confined compression. Three cases are shown: elastic contact ($\lambda_p = 0$), almost rigid

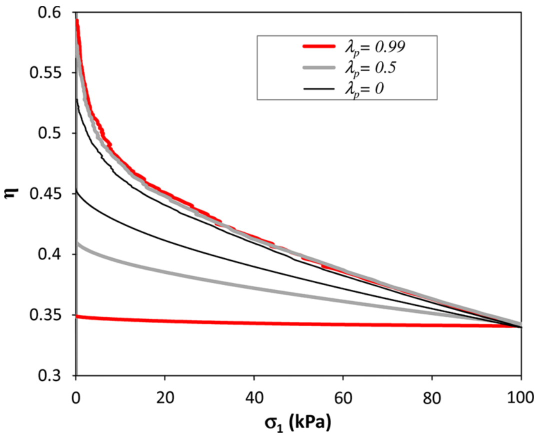

Fig. 13 Confined compression-porosity (η) versus axial strain (ε_a) with $\lambda_p = 0.99$

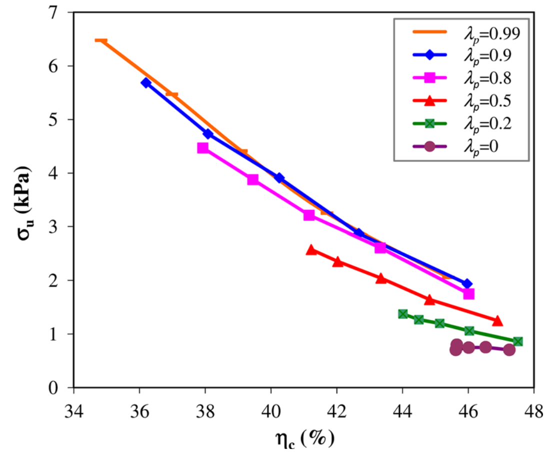

Fig. 14 Unconfined strength (σ_u) as a function of consolidated porosity (η_c) at different contact plasticity

plastic contact ($\lambda_p = 0.99$) and an intermediate value of plasticity ($\lambda_p = 0.5$). Even for elastic contact, significant bulk plastic deformation can arise from particle rearrangement. As contact plasticity increases, plastic contact deformation increases under loading resulting in a stiffer response on unloading. The softer loading responses with increasing λ_p appears surprising at first since the loading parameter k_1 was set to be constant for all three cases. The answer lies in the larger initial sample porosity when λ_p is larger. A closer look at the DEM results show that a larger contact plasticity λ_p gives rise to a greater degree of clustering during the filling process which resulted in larger voids between the clusters: this gives rise to a greater initial porosity and hence a softer response during compression. This is further highlighted in Fig. 13, where the variation in sample porosity with consolidation stress is plotted. At low consolidation stresses there is a significant difference in the observed porosities which converge onto a single loading curve at greater consolidation stresses (say 5 kPa). For the unloading path, particle contact plasticity significantly affects the bulk unloading stiffness as expected. As the contact plasticity decreases the assembly rebounds to a higher consolidated porosity.

Next the relationship between porosity and unconfined yield strength is explored. Figure 14 plots the unconfined yield strength (σ_u) against the consolidated porosity after unloading (η_c) for different contact plasticity λ_p. For each λ_p, σ_1 varies from 20 kPa to 100 kPa, with the porosity at 20 kPa being the highest in all cases. As the consolidation stress increases, the unconfined yield strength increases whilst the porosity decreases, similar to the findings from conventional soil/powder consolidation experiments [73]. The dependence on contact plasticity is evident where increasing λ_p has resulted in an increasing range of consolidated porosity η_c which gives rise to an increasing range of unconfined strength. However the lines are unique below a λ_p of 0.8, so the bulk porosity alone is not sufficient to account for

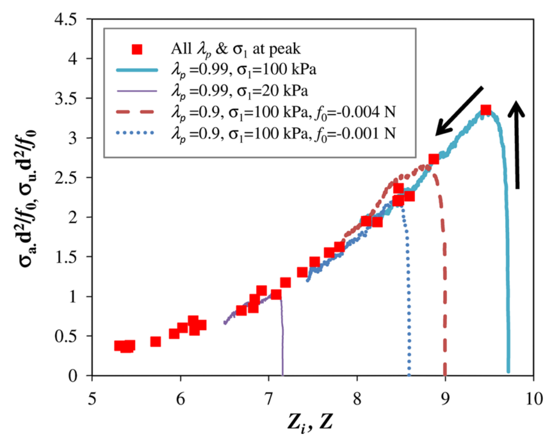

Fig. 15 Normalized unconfined strength ($\sigma_a\, d^2/f_0,\ \sigma_u\, d^2/f_0$) vs. coordination number (Z_i, Z): $f_0 = -0.002$ N unless stated explicitly

the history-dependent strength across materials with different contact plasticity. Above λ_p of 0.8 the results seem to converge. This is because as the level of contact plasticity tends above 0.8, the coordination number reaches a limiting value for a certain consolidated porosity (only true when f_o = constant and $k_{adh} = 0$).

Figure 10 above has shown how the unconfined strength increases with increasing consolidation stresses at different levels of contact plasticity. To explore the mechanism for this stress-history dependence, the stress-strain loading paths to failure for several simulations (see Fig. 7) are re-plotted in terms of the normalised axial stress and the instantaneous coordination number (Z_i) in Fig. 15. The axial stress (σ_a) is normalized with f_0/d^2 which relates to particle adhesive strength. The arrows indicate the direction of loading, from the start of unconfined loading to post-peak. As loading progresses, the axial stress increases strongly with only a very small decrease in the coordination number before reaching the peak (unconfined strength). Further loading of the sample following the peak leads to a significant drop in the observed coordination number which is associated with increasing dilation (increasing volume) [2,73] and decreasing post peak strength. This trend is consistent for all simulations in this study.

Superimposed on Fig. 15 are the normalized unconfined strength $\sigma_u d^2/f_0$ and the coordination number Z at the peak for all simulations with $k_{adh} = 0$ (denoted by ■). All data points collapse into a single 'critical curve' for the full range of contact plasticity and consolidation stresses in Fig. 10. This critical line is analogous to the concept of critical state in soil mechanics [74]. It indicates that with a constant contact adhesion f_o ($k_{adh} = 0$), the microscopic mechanism for the increasing bulk cohesion under increasing stress (stress-history effect) is due to the increasing number of contacts as a result of both the consolidation stress and contact plasticity.

7 Interaction between adhesion parameters

Cohesion in bulk materials may arise from different sources of adhesion at particle level and these are represented by two parameters in the contact model: f_0 and k_{adh}. In the section above, we explored the bulk cohesion arising only from a constant adhesive strength f_0 coupled with contact plasticity. In this section we study the interaction between the two. Simulations with constant adhesion only, load-dependent adhesion ($f_0 = 0, k_{adh} \neq 0$) only, and with both f_0 and k_{adh} not equal to zero were performed. The effect of these parameters on fill porosity, compressibility, and flow function are investigated below.

We first explore the effect of contact adhesion on fill porosity (η_f) which is defined as the sample porosity at a very low nominal consolidation stress of 0.3 kPa after filling. The fill porosity depends on the method of filling which is described in Sect. 4. While comparing the effect of k_{adh} and f_o on the filled porosity it is important to note that f_o is an adhesive strength (N) whilst k_{adh} is adhesive stiffness (N/m). It is therefore difficult to compare the effect of these parameters on the filled porosity directly.

Figure 16 shows the fill porosity arising from two different scenarios: varying f_0 with $k_{adh} = 0$ and varying k_{adh} with $f_0 = 0$. For the system with zero adhesion ($k_{adh} = 0$, $f_0 = 0$), the porosity is 41 % (as shown by dashed line in the figure) which is consistent with the findings for cohesionless paired non-spherical particle with aspect ratio of 1.5 [43]. This can be compared with the porosity of 36 % for random packed mono-disperse frictionless spheres, showing the effect of non-sphericity and friction on porosity. As the contact adhesion increases either by increasing k_{adh} or f_0, the filled porosity also increases. Contact adhesion causes the particles to stick together during the filling process and form local clusters leading to chain like structures and thus higher porosity. The adhesive forces provide a higher resistance to counteract the effect of gravity force and provide mechanical stability. These forces restrict the relative movement between particles and significantly reduce the densification due to rolling and sliding between particles [75].

The porosity initially increases slowly with increasing adhesion parameters, before it increases rapidly after reaching the inflection point and finally reaches a plateau. It should be noted that the adhesion strength at contact is not fully mobilized during the filling process. For a fair comparison of the effect of adhesion parameters on the porosity would require calculation of average mobilized adhesive (tensile) force in each system. The average tensile force (f_{at}) is defined as the ratio of total tensile force to the number of tensile contacts in the system. The Fig. 17 shows the result in Fig. 16 re-plotted in terms of average tensile force (f_{at}) normalized by gravitational force (f_g). It is notable to find that for both adhesion parameters, the interparticle force ratio

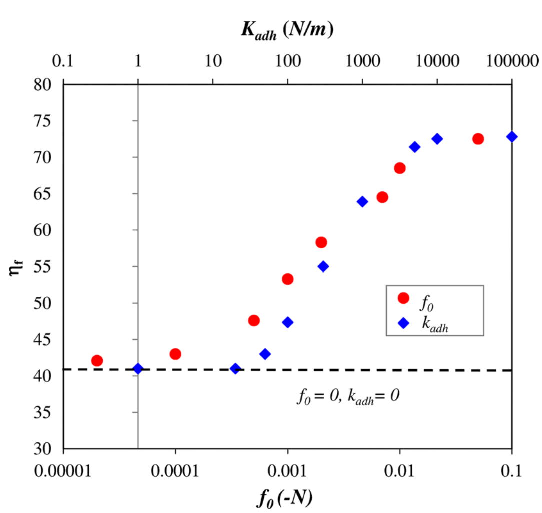

Fig. 16 Variation of porosity (η_f) at end of filling with k_{adh} and f_0 : $k_1 = 1$ and $k_2 = 10$ kN/m

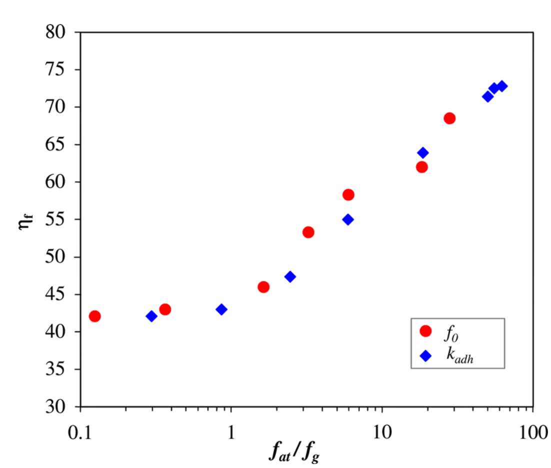

Fig. 17 Interparticle force ratio versus fill porosity

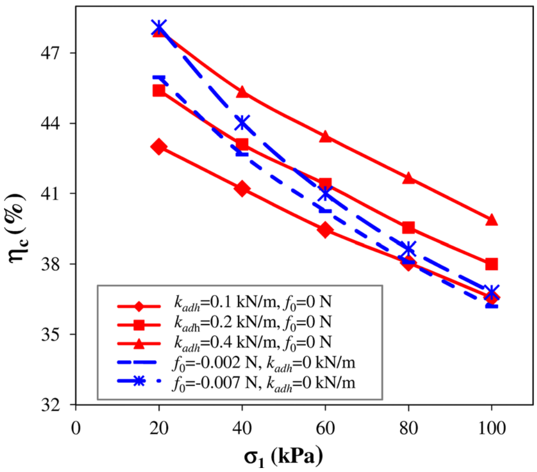

Fig. 18 Consolidated porosity (η_c) as a function of consolidation stress (σ_1): $k_1 = 1$ and $k_2 = 10$ kN/m

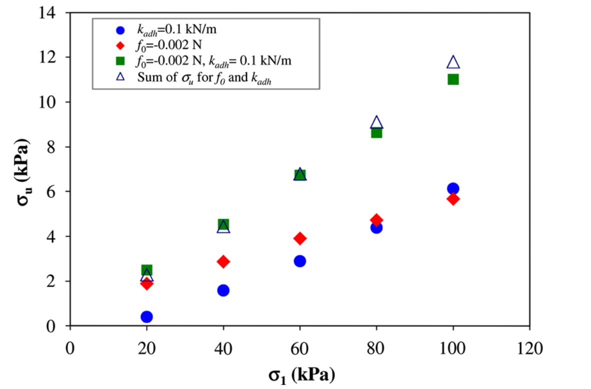

Fig. 19 Predicted flow function for different k_{adh} and f_0 with $k_1 = 1$ and $k_2 = 10$ kN/m

(f_{at}/f_g) versus porosity relationship converges to a single line. This suggests that porosity relates strongly to mobilised adhesive (tensile) force regardless of constant adhesion or load dependent adhesion.

Next we explore the effect of adhesion on bulk compressibility. The consolidated bulk porosity of the sample was calculated from the height of the consolidated sample after unloading. Figure 18 shows the consolidated porosity as a function of the consolidation stress for different adhesion parameters. As the consolidation stress increases, the consolidated bulk porosity reduces for all adhesion parameters investigated. The porosity of the sample increases as the level of adhesion (coming from f_o and k_{adh}) increases for the same consolidation stress. It is noted that the rate of decrease in porosity for k_{adh} is noticeably slower than those with a nonzero f_0 because the adhesive force is proportional to k_{adh} and is thus higher at a higher stress level, making the sample more difficult to compact. Additionally, for constant adhesion, porosity decreases more rapidly on the first application of stress and then slows down as the stress increases further. The rapid decrease in porosity at lower stress can be attributed to higher particle rearrangement resulting from the looser packing formed.

Finally, we investigate the effect of adhesion on the computed flow function. Figure 19 shows that the unconfined compressive strength increases with the consolidation stress for all the adhesion parameters, showing the stress history dependency phenomenon. Within the range of the consolidation stress studied, the predicted unconfined strength for cases with both non-zero k_{adh} and f_0 is found to be approximately the sum of the contributions from k_{adh} and f_0 separately when all the other DEM input parameters are kept constant. In this example, the slope of the flow function for the case with $k_{adh} = 0.1$ kN/m and $f_0 = 0$ N is greater compared to that with $k_{adh} = 0$ kN/m and $f_0 = -0.002$ N. In the former case, the unconfined strength increases because both the coordination number Z and the load-dependent adhesion increases with loading whilst in the latter case, the strength only increased as a function of increasing Z.

It is evident from the above discussion that the consolidated porosity (η_c) and thus the coordination number (Z) play a pivotal role in characterising the bulk cohesion. Here we explore how the mesoscopic contact parameters relate to microscopic cohesion. Rumpf [76] was the first to propose a simple model relating tensile strength (σ_t) to average adhesive strength (F_{at}) at the particle level for a system of hard mono-disperse spheres with a random isotropic packing as follows:

$$\sigma_t = \frac{F_{at}(1-\eta)Z}{\pi d^2} \tag{18}$$

where d is the diameter of particle, η is the porosity, and Z is the coordination number. It should be noted that this equation was developed for an ideal packing of hard spheres with isotropic and homogenous distribution of stresses. However, our experimental setup of uniaxial compaction is anisotropic in nature. The effect of anisotropy has been investigated by Quintanilla et al. [77] and they reported that the anisotropy does not affect the relationship significantly.

Rumpf derived the equation for tensile strength of agglomerates, whilst in this study, our focus is on the bulk compressive strength. The relationship between tensile strength and

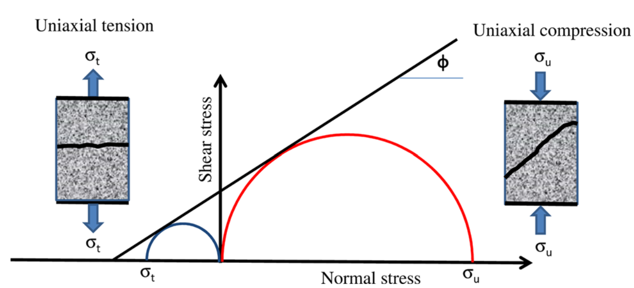

Fig. 20 Mohr circle for uniaxial tension and compression

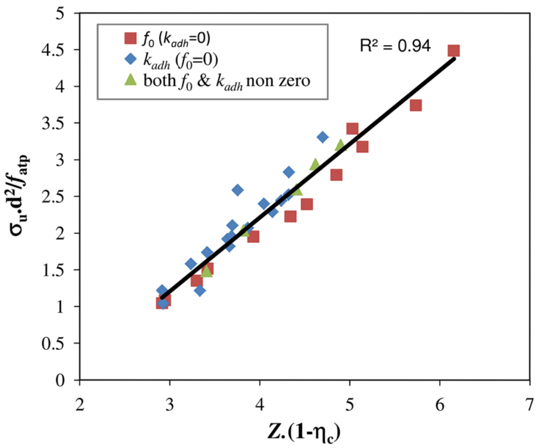

Fig. 21 Normalized unconfined compressive force ($\sigma_u d^2/f_{atp}$) as a function of coordination number (Z) and consolidated porosity (η_c)

unconfined compressive strength can be derived from the construction of Mohr Circle (see Fig. 20) by assuming a linear cohesive-frictional material. The unconfined compressive strength σ_u is thus related to the unconfined tensile strength σ_t by the simple expression:

$$\sigma_u = \frac{1+\sin\phi}{1-\sin\phi}\sigma_t \tag{19}$$

where, ϕ is angle of internal friction.

It is thus proposed that the compressive strength would have the same micromechanical origins as the tensile strength. From Eqs. 18 and 19, the normalized unconfined strength can be defined as:

$$\frac{\sigma_u}{(f_{atp}/d^2)} \propto (1-\eta_c)Z \tag{20}$$

where f_{atp} is the average tensile contact force evaluated for the whole particle system at the unconfined strength (σ_u) state. The relationship between the normalized unconfined strength and the product of Z and solid volume fraction $(1-\eta_c)$ for the full range of adhesion parameters is shown in Fig. 21. A unique linear relationship exists between the two quantities, which is analogous to the finding from Rumpf [76]. This proposes that the bulk unconfined strength for the same size particle is governed by the contact tensile force at failure, the coordination number and the solid fraction. This relationship reveals an important microstructural mechanism for bulk cohesion and can be used to facilitate unifying the characterisation and modelling of cohesive granular materials with different adhesion parameters.

It may first seem counter-intuitive why the unconfined compressive strength should relate to the tensile contact force. The effect of frictional force can be elucidated by the example simulations shown in Fig. 22. The figure shows the stress strain result during unconfined compression. Two pairs of simulations with the same simulation parameters

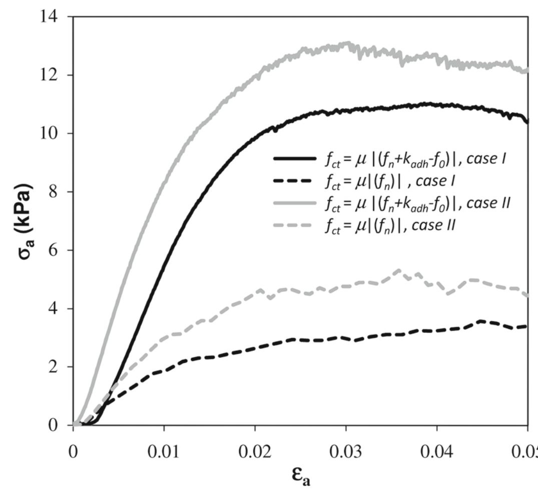

Fig. 22 Unconfined stress strain behaviour as a function of limiting friction (case I: $f_0 = -0.002\,\text{N}$, $k_2 = 10$ and $K_{adh} = 0.1\,\text{kN/m}$, case II: $f_0 = -0.007\,\text{N}$, $k_2 = 10$ and $k_{adh} = 0\,\text{kN/m}$)

but different limiting friction criteria $f_{ct} \leq \mu F_n$ and $f_{ct} \leq \mu|(f_n + k_{adh}\delta^n - f_o)|$ were conducted: the former has been explored elsewhere [13] and the latter was adopted in the present model (see Eq. 8). It can be seen that there is a huge reduction in the computed unconfined strength as well as initial stiffness if the limiting tangential force does not include the tensile strength component. This can be seen for both cases: combination of constant adhesion and load dependent adhesion (f_o & k_{adh}—case I) and constant adhesion (f_0—case II). When friction limit does not include tensile (adhesive) component, the tensile force remains similar but the unconfined strength reduces significantly due to reduced shearing resistance. This confirms that the contribution of adhesive force to the limiting friction has a significant effect on the bulk unconfined strength. This is also in line with the observation that under uniaxial unconfined compression, the mode of failure is predominantly shear failure rather than tensile failure [73,74].

8 Conclusions

DEM simulations and micromechanical analysis of cohesive powders using an adhesive elasto-plastic contact model have been presented. The model is capable of capturing the important stress-history dependency of powders' unconfined cohesive strength, as observed in experiments. This suggests that the elasto-plastic adhesive model may be used to simulate cohesive solids subjected to different flow and stress regimes.

The particle contact plasticity has been found to be essential for capturing the stress history dependence and to produce a realistic flow function. Micromechanical analysis revealed that increased particle contact plasticity increases the bulk plasticity. The contact plasticity prevents excessive elastic rebound at the contact level which leads to a lower porosity on the application of stress. For constant adhesive strength, the unconfined strength has been shown to correlate with the instantaneous coordination number at the peak state. This provides a microscopic explanation for the unconfined strength variation and also indicates that the coordination number can be used as a state variable to avoid invoking the history effect. In some sense, this correlation is a new microscopic (or meso-scale) flow function.

When the contact adhesive strength increases, the filled porosity also increases. Higher adhesive forces allow the particles to stick together during the filling process, leading to stronger chain like structure and ultimately higher filled porosity. The adhesive forces provide a high resistance to counteract the effect of gravity force and provide some mechanical stability which restricts the relative movement between particles. For both load dependent k_{adh} and constant adhesive strength f_0, a unique relationship between porosity and mobilized adhesive force was found. On the application of confined compression, the load-dependent adhesion provides a greater resistance to volumetric compression because the adhesive strength is proportional to k_{adh} and is thus higher at higher stress levels, making the sample more difficult to compact.

While comparing the effect of adhesion parameters on unconfined strength within a range of consolidation stresses, it has been found that the predicted unconfined strength for cases with both non-zero k_{adh} and f_0 is approximately the sum of the contributions from k_{adh} and f_0 separately; when all the other DEM input parameters are kept constant. A linear relationship has been established between the normalized unconfined strength and the product of coordination number and solid volume fraction. This gives a general microscopic (or meso-scale) flow function for different materials with distinct cohesion strength origins.

It has been found that contribution of adhesive force to the limiting friction has a significant effect on bulk unconfined strength. Failure to include the adhesive contribution in the calculation of the frictional resistance may lead to under-prediction of unconfined strength and incorrect failure mode.

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